

AC2206 & AC3302

Investment Appraisal

Winter 2023 Examination

Paper 1 · 1.5 hours · Answer Question 1 and one other question · Total 100 Marks

Module	AC2206 & AC3302 — Investment Appraisal
Session	Winter 2023
Paper	1 of 1
External Examiner	Professor Thomas O'Connor
Internal Examiner	Mr. David Humphreys
Structure	Question 1 (Compulsory, 50 Marks) + one of Question 2 or Question 3 (50 Marks)
This scheme covers	Question 1, Question 2 and Question 3 in full, so you can study either route through the paper

Own-figure rule. Where a candidate makes an error early in a multi-step question, full method marks are still awarded for correctly applying the required technique to their own, incorrect figure in every subsequent step. This applies throughout every question in this scheme.

Quality check. Every calculation in this scheme has been independently verified and cross-checked for accuracy before being written up.

Question 1 (Compulsory)

50 Marks — NPV under Uncertainty, Risk & Capital Rationing

Scenario

Augusta plc has discovered a gold deposit in Ireland. On 1 January 2023 the firm is deciding whether to commence extraction. Equipment cost **€3 million** and a government licence fee of **€2 million** are both payable on 1 January 2023. The Irish government takes **25% of all revenues**. Production delays mean the gold will not be ready for sale until 1 January 2024. Two independent uncertainties affect annual cash sales:

Quantity of gold	Probability
High	0.4
Low	0.6

Demand scenario	Quantity of gold found	Annual cash sales
Strong Demand (prob. 0.6)	High	€15 million
	Low	€5 million
Weak Demand (prob. 0.4)	High	€9 million
	Low	€3 million

Gold is extracted over a **5-year period**, with 100% advance payment received on 1 January of each year. Required rate of return: **10%**.

(a) Required

30 Marks

Calculate the expected net present value of the investment project.

FULL MODEL ANSWER

Quantity of gold (a geological uncertainty) and demand (a market uncertainty) are independent, so the probability of each of the four combined scenarios is the product of the two individual probabilities:

Scenario	Probability	Annual Cash Sales	Prob. × Sales
Strong Demand & High Quantity	$0.6 \times 0.4 = 0.24$	€15m	€3.600m
Strong Demand & Low Quantity	$0.6 \times 0.6 = 0.36$	€5m	€1.800m
Weak Demand & High Quantity	$0.4 \times 0.4 = 0.16$	€9m	€1.440m
Weak Demand & Low Quantity	$0.4 \times 0.6 = 0.24$	€3m	€0.720m
Expected annual gross revenue	1.00		€7.560m

The Irish government takes 25% of all revenues, so Augusta retains 75%:

$$\text{Expected net annual cash flow} = \text{€7.560m} \times 75\% = \text{€5.670m}$$

Modelling assumption, stated explicitly. As with the equivalent question in other papers in this series, this scheme treats the quantity and demand outcomes as resolved once (a single combined scenario applies for the life of the project) and uses the single probability-weighted expected annual cash flow as a level amount received in every one of the 5 years, consistent with the fixed € annual cash sales figures given per scenario.

Question 1 (continued)

Both the equipment cost and the licence fee are paid on 1 January 2023 ($t = 0$). Because of production delays, the first advance payment is received on 1 January 2024 — exactly one year later, at $t = 1$ — making this a standard 5-year ordinary annuity ($t = 1$ through $t = 5$).

Timing	Cash Flow	PV Factor @ 10%	Present Value
$t = 0$ (1 Jan 2023)	Equipment (€3m) + Licence fee (€2m) = –€5.000m	1.000	–€5.000m
$t = 1$	+€5.670m	0.909	€5.154m
$t = 2$	+€5.670m	0.826	€4.683m
$t = 3$	+€5.670m	0.751	€4.258m
$t = 4$	+€5.670m	0.683	€3.873m
$t = 5$	+€5.670m	0.621	€3.521m
Expected NPV			€16.49m

Cross-check. Using the 5-year, 10% PV of annuity factor from the table (3.791) directly: $NPV = -€5.000m + (€5.670m \times 3.791) = -€5.000m + €21.495m = \text{€16.49m}$, matching the year-by-year calculation above.

Mark Allocation — Part (a), 30 Marks

Tier	Marks	What separates this tier
24-30	24-30	Correct joint probability weighting of the 2×2 scenario table, correct 75%-retention adjustment for the government's 25% share, correct identification of the annuity starting at $t=1$, and correct final NPV with clear working.
16-23	16-23	Correct overall method but one component error — e.g. government share applied to net cash flow rather than gross revenue, or annuity timing treated as starting at $t=0$ (own-figure carried through).
8-15	8-15	Expected annual cash flow correctly identified but the government revenue share and/or discounting applied incorrectly.
0-7	0-7	Joint probability structure not recognised, or no credible NPV attempt.

Question 1 (Compulsory)

50 Marks — NPV under Uncertainty, Risk & Capital Rationing

(b) Required

10 Marks

Outline and explain three potential risks associated with this project. What steps could Augusta plc take to mitigate these risks?

FULL MODEL ANSWER

1. Geological / extraction (quantity) risk: The actual quantity of gold recoverable may differ from the High/Low estimates — geological surveys are inherently uncertain, and the site could yield materially less gold than forecast. **Mitigation:** commission further independent geological surveys and appraisal drilling before committing capital, and consider a phased investment approach.

2. Commodity price / demand risk: Annual cash sales depend heavily on prevailing gold demand and price conditions, driven by global macroeconomic factors (e.g. investor demand for gold as a safe-haven asset) outside Augusta's control, which could deteriorate further than the 'Weak Demand' scenario modelled. **Mitigation:** use derivatives such as gold futures or forward sales contracts to lock in a minimum price for a portion of expected production.

3. Political / regulatory risk: The Irish government's 25% revenue share is a current policy setting that could change over the life of the project, directly reducing Augusta's net cash flows. **Mitigation:** seek to negotiate contractual/fiscal stability clauses with the government at the outset of the licence agreement, and run sensitivity analysis on the government's share to understand the project's exposure to this risk.

Other acceptable risks: Candidates may instead or additionally discuss production delay risk, environmental/regulatory compliance risk, currency risk (gold is typically priced in USD), or interest rate risk affecting the discount rate — any three well-explained risks with a genuine, relevant mitigation each should receive full credit.

Mark Allocation — Part (b), 10 Marks

Tier	Marks	What separates this tier
8-10	8-10	Three distinct, well-explained risks specific to this project, each paired with a genuine, relevant mitigation strategy.
5-7	5-7	Three risks identified but explanations generic, or mitigation steps vague.
2-4	2-4	One to two risks identified with reasonable explanation.
0-1	0-1	No meaningful discussion of project risk.

Question 1 (Compulsory)

50 Marks — NPV under Uncertainty, Risk & Capital Rationing

(c) Required

10 Marks

In the context of investment appraisal, outline the causes of and the effects of both hard and soft capital rationing for companies.

FULL MODEL ANSWER

Capital rationing arises when a company faces a constraint on the total funds available for investment, forcing it to choose between positive-NPV projects rather than accepting all of them.

	Hard Capital Rationing	Soft Capital Rationing
Source of constraint	Externally imposed, by lenders/capital markets	Internally imposed, by the company's own management
Typical causes	Lenders unwilling to extend further credit; restrictive loan covenants; poor credit rating; depressed capital market conditions limiting new equity issuance	Management's desire to maintain a conservative capital structure; decentralised budgeting with fixed divisional ceilings; a deliberate policy to filter out all but the strongest projects; reluctance to dilute ownership/control via new equity
Effects	Positive-NPV projects may have to be rejected purely due to lack of access to external funds, potentially destroying shareholder value	Positive-NPV projects may still be rejected, but the constraint is self-imposed and can in principle be relaxed by management; often reflects a deliberate strategy of encouraging more careful project selection or preserving financial flexibility

Common consequence — ranking under rationing: Under either form of rationing, a company facing a fixed capital budget should, in a single-period setting, rank divisible projects by their Profitability Index (PV of future cash flows ÷ initial investment) rather than by NPV alone, since this best identifies which combination of projects maximises total NPV given the limited capital available.

Candidates who correctly distinguish the external (hard) versus internal (soft) source of the constraint, give at least two plausible causes for each, and identify the common effect (forgone positive-NPV projects) alongside the specific nuance for soft rationing (self-imposed and reversible) should receive full credit.

Mark Allocation — Part (c), 10 Marks

Tier	Marks	What separates this tier
8-10	8-10	Clear distinction between hard and soft rationing with correct causes for each and a sound discussion of effects, including the nuance that soft rationing is self-imposed.
5-7	5-7	Hard vs soft distinction made but causes or effects for one type underdeveloped.
2-4	2-4	Some understanding shown but hard and soft rationing conflated.
0-1	0-1	No meaningful discussion of capital rationing.

Question 2

50 Marks — Choice question: answer EITHER Question 2 OR Question 3

How to use this scheme. The paper requires Question 1 plus **one** of Question 2 or Question 3 — both are marked out of 50 and count equally. This scheme covers both in full so you can revise whichever route you are more comfortable with.

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Question 2

50 Marks — Choice question: answer EITHER Question 2 OR Question 3

Scenario

Baltusrol plc is considering new plant and machinery costing **€20 million** (January 2023) to produce a new electric drill over the next 5 years, with zero scrap value at the end. Budgeted at 200,000 units/year, price €200/unit, operating cost €140/unit (both figures for year 1). Corporation tax 12.5%, with a **one-year delay** in tax payment. Capital allowances: 40% in year 1, then 25% reducing balance. Required rate of return 9%; inflation on both price and costs is 5% p.a.

(a) Required

30 Marks

Using the NPV method of project appraisal, recommend whether or not the project should be accepted.

FULL MODEL ANSWER

The year-1 price and cost figures are the starting point, and since both inflate at the identical 5% rate, the per-unit contribution margin also grows at exactly 5% per year:

Year	Price/unit	Cost/unit	Margin/unit	Volume	Annual Contribution
1	€200.00	€140.00	€60.00	200,000	€12.000m
2	€210.00	€147.00	€63.00	200,000	€12.600m
3	€220.50	€154.35	€66.15	200,000	€13.230m
4	€231.53	€162.07	€69.46	200,000	€13.892m
5	€243.10	€170.17	€72.93	200,000	€14.586m

Capital allowances (40% year 1, then 25% reducing balance, balancing allowance in year 5)

Year	Opening TWDV	Allowance	Closing TWDV
1	€20.000m	40% × 20.000 = €8.000m	€12.000m
2	€12.000m	25% × 12.000 = €3.000m	€9.000m
3	€9.000m	25% × 9.000 = €2.250m	€6.750m
4	€6.750m	25% × 6.750 = €1.688m	€5.063m
5 (disposal, nil proceeds)	€5.063m	Balancing allowance = €5.063m	€0.000m
Total capital allowances		€20.000m	

Total capital allowances claimed over the 5 years (€20.000m) exactly equal the original cost of the asset, confirming the year-5 balancing allowance (equal to the remaining tax written-down value, since scrap proceeds are nil) has been correctly included.

Question 2 (continued)

Taxable profit and tax (12.5%, paid one year in arrears)

Year	Contribution	Less: Capital Allowance	Taxable Profit	Tax @ 12.5%	Tax paid in Year
1	€12.000m	€8.000m	€4.000m	€0.500m	2
2	€12.600m	€3.000m	€9.600m	€1.200m	3
3	€13.230m	€2.250m	€10.980m	€1.373m	4
4	€13.892m	€1.688m	€12.204m	€1.526m	5
5	€14.586m	€5.063m	€9.524m	€1.190m	6

NPV calculation

Year (t)	Contribution CF	Tax Paid	Net Cash Flow	PV Factor @ 9%	Present Value
0	—	—	–€20.000m	1.000	–€20.000m
1	€12.000m	€0.000m	€12.000m	0.917	€11.004m
2	€12.600m	€0.500m	€12.100m	0.842	€10.188m
3	€13.230m	€1.200m	€12.030m	0.772	€9.287m
4	€13.892m	€1.373m	€12.519m	0.708	€8.863m
5	€14.586m	€1.526m	€13.061m	0.650	€8.489m
6	—	€1.190m	–€1.190m	0.596	–€0.709m
NPV					+€27.12m

Recommendation. The project has a strongly positive expected NPV of approximately **+€27.1 million**, comfortably exceeding the €20 million initial outlay. **Baltusrol plc should accept the project.** As with the equivalent question in other papers in this series, note the terminal tax cash outflow in year 6 — tax on year 5's profit is paid the year after the project's operating cash flows end, and must not be omitted from the NPV even though no contribution is received that year.

Mark Allocation — Part (a), 30 Marks

Tier	Marks	What separates this tier
24-30	24-30	Correct inflated contribution margins, correct capital allowance schedule including the year-5 balancing allowance, correct one-year tax lag (including the year-6 terminal tax outflow), correct NPV and a clear accept/reject recommendation.
16-23	16-23	Correct overall structure with one component error — e.g. inflation applied to only price or only cost, or the year-6 terminal tax payment omitted (own-figure carried through).
8-15	8-15	Contribution and capital allowances calculated but tax timing not correctly applied, or capital allowances not correctly reduced by the balancing allowance treatment.
0-7	0-7	No credible attempt to incorporate tax and capital allowances into the cash flow.

Question 2

50 Marks — Choice question: answer EITHER Question 2 OR Question 3

(b) Required

20 Marks

Outline the main findings of the empirical investigations of the investment appraisal process. To what extent does the real world practice of investment appraisal deviate from the theory you have studied?

FULL MODEL ANSWER

A substantial body of survey research (e.g. Pike 1996 in the UK; Graham & Harvey 2001 in the US; Arnold & Hatzopoulos 2000 in the UK) has examined how companies actually appraise investment projects in practice, and the findings show both convergence with, and notable departures from, the textbook theory.

DCF technique adoption has risen substantially over time: Earlier surveys found relatively low usage of NPV and IRR, but more recent surveys consistently find that the large majority of large firms now use NPV and/or IRR as a primary appraisal technique, reflecting greater sophistication in corporate finance practice.

Payback period remains persistently popular despite its theoretical shortcomings: Even where NPV/IRR are used, payback period is frequently used alongside them, despite ignoring the time value of money in its simple form and ignoring cash flows beyond the payback point — a clear and persistent deviation from pure theory, often explained by its simplicity and intuitive appeal.

Firms typically use multiple techniques together, not a single 'correct' method: Most firms apply a suite of criteria (e.g. NPV, IRR, payback, and accounting rate of return) simultaneously, using each to inform different aspects of the decision (value creation, risk, and reported performance impact respectively).

Firm size and sophistication matter: Larger firms, and firms with more financially qualified management, are significantly more likely to use formal DCF techniques; smaller firms disproportionately rely on payback period and rules of thumb.

Discount rate practice deviates from theory: Finance theory calls for project-specific, risk-adjusted discount rates, but survey evidence consistently finds many firms apply a single company-wide hurdle rate (often based on WACC) across all projects regardless of individual project risk.

Formal treatment of uncertainty is limited in practice: While the theory offers tools such as decision trees, Monte Carlo simulation and real options analysis, survey evidence finds these formal techniques are used by only a minority of firms; sensitivity and scenario analysis are far more commonly used in practice.

Capital rationing and strategic/qualitative overlays are common:

Consistent with the discussion of hard and soft capital rationing in Question 1(c), many firms operate under a capital constraint, and final project selection often reflects strategic fit and qualitative judgement alongside — sometimes ahead of — the numerical NPV/IRR output.

Candidates who correctly identify at least three to four of these deviations (rising but incomplete DCF adoption, persistent payback usage, single hurdle rates versus risk-adjusted rates, and limited use of formal uncertainty techniques) with reference to the general thrust of the survey literature should receive full credit — exact citation of specific studies is not required, though it strengthens the answer.

Mark Allocation — Part (b), 20 Marks

Tier	Marks	What separates this tier
16-20	16-20	Clear, well-developed discussion of at least four distinct empirical findings, explicitly addressing how practice deviates from theory, ideally with reference to specific research.
11-15	11-15	Good discussion of two to three findings; deviation from theory addressed but not always explicitly drawn out.
6-10	6-10	Techniques described but with little or no discussion of survey evidence or the theory-practice gap.
0-5	0-5	No meaningful discussion of empirical investment appraisal practice.

Question 2 Total: 50 Marks

Question 3

50 Marks — Choice question: alternative to Question 2

(a) Required

20 Marks

An investor believes that shares in Bank of Ireland are currently underpriced at €9.50 and that they will trade significantly higher in three months' time. The premium on a three month Bank of Ireland option with an exercise price of €10 is currently €1.

- Indicate the option strategy the investor would employ to take advantage of any share price increase.
- Calculate the profit/(loss) that would accrue from such a strategy if the Bank of Ireland share price after three months was: (a) €9 (b) €14. Supplement your answer with appropriate payoff diagrams.

FULL MODEL ANSWER

(i) Strategy: The investor should buy a call option on Bank of Ireland with a €10.00 exercise price (a **long call**). This gives the right, but not the obligation, to buy the shares at €10.00 — since the investor expects the price to rise significantly, the call captures that upside for a much smaller upfront outlay (the €1.00 premium) than buying the shares outright.

(ii) Profit/loss on the call option, per share, at the option's three-month expiration (matching the time horizon given, so this is a straightforward expiration payoff with no timing inconsistency to flag):

$$\text{Profit} = \text{Max}(\text{Share Price} - \text{Exercise Price}, 0) - \text{Premium Paid}$$

Share Price	Call exercised?	Intrinsic Value	Less: Premium Paid	Profit/(Loss) on Call
€9.00	No — price below strike	€0.00	€1.00	(€1.00)
€14.00	Yes — buy at €10 vs market €14	€4.00	€1.00	€3.00

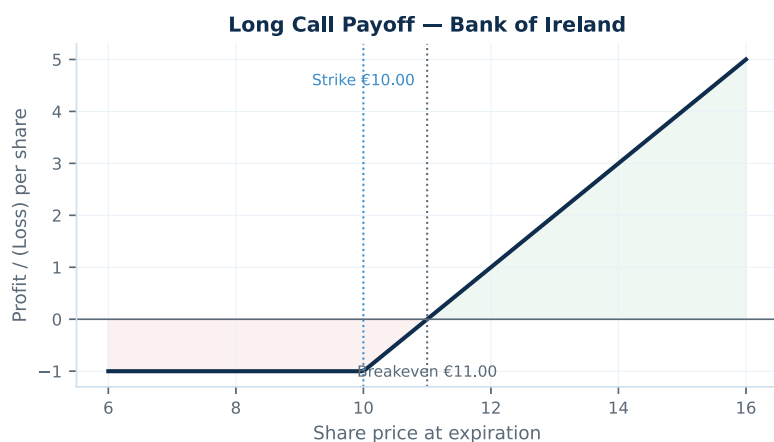


Fig. 1 — Long call payoff, Bank of Ireland (strike €10.00, premium €1.00, breakeven €11.00)

At €9.00 the option expires worthless (below the €10.00 strike), so the loss is capped at the €1.00 premium — the defining capped-downside, uncapped-upside profile of a long call. At €14.00 the option is comfortably in the money, delivering a €3.00 profit per share on a €1.00 outlay. Breakeven is $€10.00 + €1.00 = €11.00$.

Mark Allocation — Part (a), 20 Marks

Tier	Marks	What separates this tier
16-20	16-20	Correct strategy (long call) with clear reasoning; both profit/loss figures correct; payoff diagram correctly shaped and labelled (strike, breakeven).
11-15	11-15	Correct strategy and both figures correct, but diagram missing or materially incorrect.
5-10	5-10	Correct strategy identified but one or both profit/loss figures wrong, or a put option used instead of a call.
0-4	0-4	Wrong strategy or no credible attempt.

Question 3

50 Marks — Choice question: alternative to Question 2

(b) Required**20 Marks**

'A straddle should be purchased if an investor believes that asset prices are likely to remain in a narrow range.' Briefly comment on the above statement. Outline the payoff to a straddle strategy in the event that asset prices were to move in a wide range.

FULL MODEL ANSWER

The statement in the question is incorrect.

What a long straddle actually is: A long straddle involves buying both a call option AND a put option on the same underlying asset, with the same strike price and the same expiration date. The combined cost is the sum of the two premiums, and the position profits if the underlying moves far enough in EITHER direction — up or down — to more than cover that combined premium.

Why the statement is wrong: An investor who believes prices will remain in a **narrow** range should NOT buy a straddle. If the price stays close to the strike at expiration, both the call and the put expire worthless (or close to it), and the investor loses the full combined premium paid — the worst possible outcome for a long straddle occurs precisely when the underlying stays in a narrow range. A long straddle is instead the appropriate strategy for an investor who expects a **large** price move but is **uncertain of the direction** — for example, ahead of a binary event such as an earnings announcement or a regulatory decision.

The correct strategy for a narrow-range view: An investor who genuinely believes prices will stay in a narrow range would instead be better served by **writing (selling) a straddle** — collecting the combined premium upfront and profiting if the price does indeed stay close to the strike, since both options would then expire worthless. This carries the opposite (unlimited) risk profile if the investor's narrow-range view turns out to be wrong.

Question 3 (continued)

Payoff to a long straddle if prices move in a wide range

Take an illustrative example based on the Bank of Ireland scenario above: a strike price of €10.00, with a call premium of €1.00 and — for illustration, since no put premium is given in this question — an assumed put premium of €1.00, giving a combined premium of €2.00.

$$\text{Profit} = \text{Max}(S - K, 0) + \text{Max}(K - S, 0) - \text{Combined Premium}$$

Share Price at Expiry	Call Payoff	Put Payoff	Less: Combined Premium	Total Profit/(Loss)
€5.00 (wide move down)	€0.00	€5.00	€2.00	€3.00
€10.00 (no move — worst case)	€0.00	€0.00	€2.00	(€2.00)
€15.00 (wide move up)	€5.00	€0.00	€2.00	€3.00

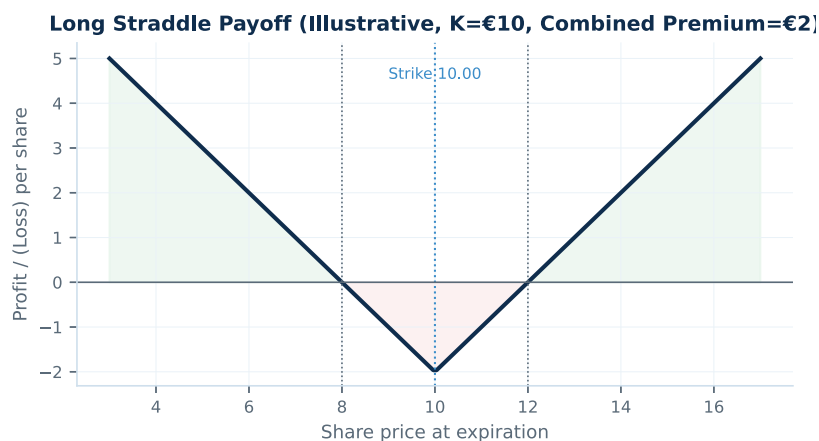


Fig. 2 — Long straddle payoff: losses are worst in a narrow range around the strike; profit grows in either direction as the price move widens

Key shape of the payoff: The payoff diagram is V-shaped, with the point of maximum loss (equal to the combined premium paid) sitting exactly at the strike price. There are two breakeven points — strike minus combined premium, and strike plus combined premium — and profit increases without limit as the price moves further from the strike in either direction, which is exactly why this strategy suits a 'big move, uncertain direction' view rather than the narrow-range view described in the original statement.

Mark Allocation — Part (b), 20 Marks

Tier	Marks	What separates this tier
16-20	16-20	Correctly identifies the statement as false, with clear reasoning for why narrow-range views should NOT prompt buying a straddle; correctly identifies that a wide/uncertain-direction move is the appropriate scenario; payoff correctly outlined and diagram/shape correctly described (V-shaped, max loss at strike, two breakevens).
11-15	11-15	Correctly identifies the statement as false with reasonable explanation, but payoff description for the wide-range case incomplete.
5-10	5-10	Straddle mechanics described correctly, but fails to correctly challenge the false premise of the statement.
0-4	0-4	Statement accepted as correct, or straddle mechanics fundamentally misunderstood.

Question 3

50 Marks — Choice question: alternative to Question 2

(c) Required

10 Marks

Briefly outline the effect of a change in the risk free rate of interest on the prices of both call and put options.

FULL MODEL ANSWER

Variable	Effect on Call price	Effect on Put price
Risk-free interest rate (r)	Increases as r rises	Decreases as r rises

The cost-of-carry / present-value mechanism: A higher risk-free rate has two related effects that both point the same way for a call: it raises the opportunity cost of holding cash rather than the option (i.e. the cost of 'carrying' a position in the underlying via borrowing), and — more directly relevant here — it lowers the present value of the fixed exercise price that will be paid in the future if the call is exercised. Since a call holder effectively defers paying the strike price until (possibly) exercising later, a higher discount rate makes that deferred payment 'cheaper' in present-value terms, which increases the call's value today.

Why the effect is reversed for a put: A put holder is in the opposite position — they will potentially receive the fixed exercise price in the future by selling the underlying. A higher discount rate lowers the present value of that future fixed receipt, which reduces the put's value today. This opposite-sign relationship is captured formally in put-call parity ($C - P = S - Ke^{-rT}$): as r rises, Ke^{-rT} falls, which (holding S , K , and T fixed) must widen the gap in favour of the call and against the put.

Practical magnitude: Among the five Black-Scholes inputs, the risk-free rate typically has the smallest practical effect on option prices over realistic ranges of interest rate movement, compared to changes in the underlying price or volatility — interest rate sensitivity (formally the option's 'rho') matters more for longer-dated options, where the present-value effect of discounting the exercise price compounds over a longer horizon.

Because this question isolates a single Black-Scholes variable, full marks require going beyond a one-line statement of direction — a sound answer should explain the present-value/cost-of-carry mechanism behind the opposite-sign effects on calls and puts, not simply state that calls rise and puts fall.

Mark Allocation — Part (c), 10 Marks

Tier	Marks	What separates this tier
8-10	8-10	Correct direction of effect for both calls (increase) and puts (decrease) as r rises, WITH a sound explanation of the underlying present-value/cost-of-carry mechanism.
5-7	5-7	Correct direction of effect for both calls and puts, but with little or no explanation of the underlying mechanism.
2-4	2-4	Only one of the two directions (call or put) correctly identified.
0-1	0-1	Direction of effect incorrect or no credible attempt.

Question 3 Total: 50 Marks